

Year 10 Science (General)
Semester 1 Examination Revision 2019
Physical Sciences

1. What is the difference between **distance** and **displacement**?

DISTANCE: movement in any direction.

DISPLACEMENT: measurement from start to finish in a straight line in a particular direction.

2. Convert the following speeds to metres/second (m/s).

(a) 67 cm/s = 0.67 m/s

(b) 275 km/h $\frac{275}{3.6} = 76.4 \text{ m/s}$

(c) 35.8 mm/s 0.038 m/s

3. A man drives his car 800 m due East and then turns North, driving 300 m before stopping at a petrol station. It took him 120 s to get there.

- (a) What distance he has driven.

$$d = 800 + 300 \\ = \underline{1100 \text{ m}}$$

- (b) Calculate his average speed for the journey.

$$sp = ?$$

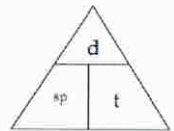
$$d = 1100 \text{ m}$$

$$t = 120 \text{ s}$$

$$sp = \frac{d}{t}$$

$$= \frac{1100}{120}$$

$$= \underline{9.17 \text{ m/s}}$$



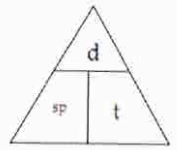
4. A balloon filled with hydrogen gas has a velocity of 40 m/s East. How long will it take to go 450 m?

$$sp = 40 \text{ m/s}$$

$$d = 450 \text{ m}$$

$$t = ?$$

$$\begin{aligned} t &= \frac{d}{sp} \\ &= \frac{450}{40} \\ &= \underline{11.2 \text{ s}} \end{aligned}$$



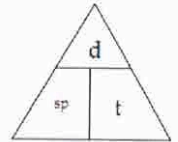
5. What is the average speed of a plane if it flies 2700 km in 2.3 hours? Give the answer in km/h.

$$sp = ?$$

$$d = 2700 \text{ km}$$

$$t = 2.3 \text{ hrs}$$

$$\begin{aligned} sp &= \frac{d}{t} \\ &= \frac{2700}{2.3} \\ &= \underline{1174 \text{ km/h}} \end{aligned}$$



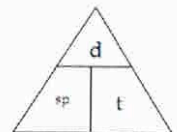
6. The driver of a car moving at 100 km/h along the freeway sees an emergency 80 m ahead and moves her foot from the accelerator to the brake. This takes 1.45 s to occur. How far has the car travelled before the driver has activated the brakes.

$$sp = 27.8 \text{ m/s}$$

$$d = ?$$

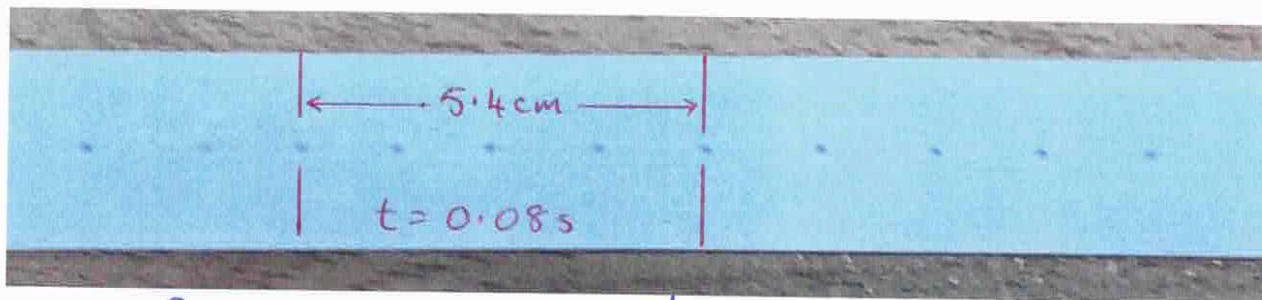
$$t = 1.45 \text{ s}$$

$$\begin{aligned} d &= sp \times t \\ &= 27.8 \times 1.45 \\ &= \underline{40.3 \text{ m}} \end{aligned}$$



Choose any 5 dots

7. The following ticker tape show the constant motion of an object in the laboratory. Determine the value of the average speed shown. Show all of your working.

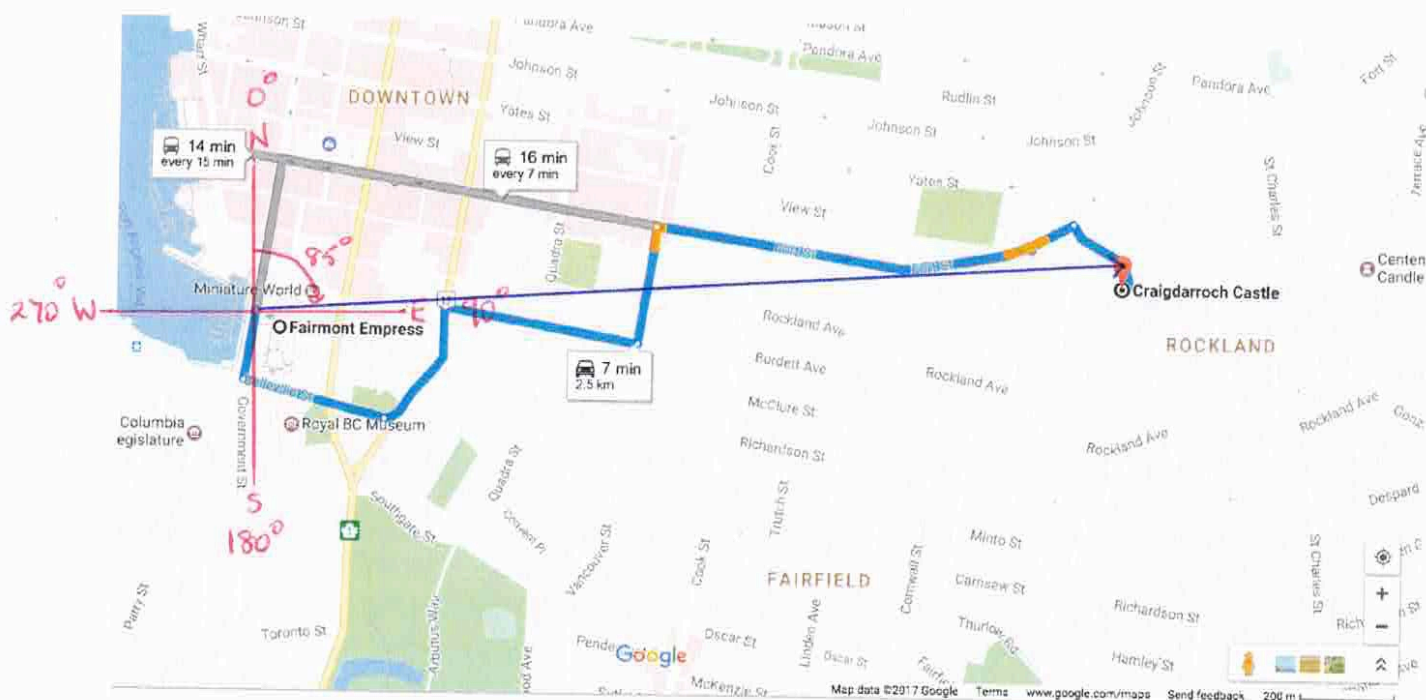


$$\begin{aligned} sp &= ? \\ d &= 0.054 \text{ m} \\ t &= 0.08 \text{ s} \end{aligned}$$

$$\begin{aligned} sp &= \frac{d}{t} \\ &= \frac{0.054}{0.08} \\ &= \underline{0.675 \text{ m/s}} \end{aligned}$$

8. Use the scale in the map below to work out the **distance and displacement** from the Fairmont Empress hotel to Craigdarroch Castle. Use either the blue or grey route.

Remember to give a bearing or direction for the displacement.



distance: 2.5 km (blue path)

displacement: 1900 m bearing 85°
(± 50 m) (± 2°)

$$1.2 \text{ cm} = 200 \text{ m}$$

$$\begin{aligned} S &= \frac{11.4}{1.2} \times \frac{200}{1} \\ &= 1900 \text{ m} \end{aligned}$$